OCR in the Foundation Model Era: From Specialist Models to Unified Models

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Abstract—Optical Character Recognition (OCR) is among the oldest and most widely deployed applications of artificial intelligence. This survey reviews its development, focusing on the Foundation Model Era. We define OCR through *faithful transcription*: within a specified visual scope, OCR should completely reconstruct the text, notation, and document structure observable in the source image, and every textual, notational, or structural element in the output must be verifiable from visual evidence in that image. We trace the development from specialist OCR models toward unified models, examining joint text detection and recognition, structured sequence generation, and the use of pretrained vision–language models (VLMs). We organize contemporary OCR systems into five coexisting families: modular toolkits and task-specific models, specialized document VLMs, hybrid workflows, general-purpose VLMs, and commercial products and managed services. We compare their architectures, data, and training, examining hallucination and omission, insufficient spatial grounding, and inference cost in generative OCR. We review evaluation from character and scene text recognition to parsing individual structural elements and full-page transcription, highlighting gaps in consistent output definitions, attribution of performance to individual components, and coverage of real capture conditions and visible content and structure. Finally, we discuss future directions for OCR, focusing on transcription reliability, structural recovery, and processing efficiency. We continuously track related work at https://github.com/zhihengli-casia/Awesome-OCR-in-the-Foundation-Model-Era.

Index Terms—Optical Character Recognition, Generative Document Parsing, Vision–Language Models.

1 Introduction

Optical Character Recognition (OCR) is among the oldest and most widely deployed applications of artificial intelligence. Efforts to read printed text by machine date back to the 1910s [1], and by the 1950s dedicated OCR devices matched optically sensed character patterns against stored templates to produce machine-readable codes [2].

Over the following decades, OCR moved well beyond its original reliance on dedicated hardware. The targets of recognition expanded along several axes: from printed to handwritten text [3], from scanned pages to text captured in natural scenes as cameras became ubiquitous [4], and from plain passages to documents with rich structures such as tables, formulas, and layouts with multiple columns [5], [6]. Generative models, increasingly built on pretrained vision–language models (VLMs), produce JSON, Markdown, HTML, or LaTeX directly from document images [7], [8].

What counts as OCR? The literature uses OCR with different boundaries: technology for scanned documents [4], camera-based OCR [9], artificial optical signals [8], text and visual structure recovery in holistic OCR [10], and multimodal evaluations that include information extraction and document question answering [11]. In this survey, we define OCR through *faithful transcription*. Within a specified visual scope, OCR aims to completely reconstruct the text, notation, and document structure observable in the source image. Every textual, notational, or structural element in the output must be verifiable from visual evidence in that image (Fig. 1). Normalization must follow a declared output convention. Our main coverage includes printed, handwritten, and scene text, mathematical formulas, tables, layout, and reading order. Chart data, flowchart connections, musical notation, and chemical structures can also be recovered from visual evidence; we discuss these as related extensions with selective coverage.

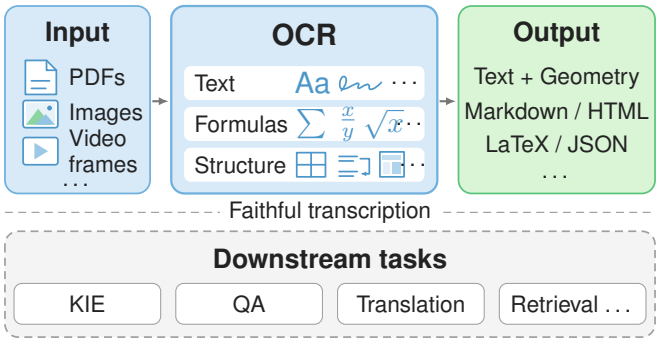

Fig. 1. **Scope of this survey.** Within a specified visual scope, OCR aims to completely recover visible text, notation, and document structure, with every output element verifiable against the source image. Main coverage spans printed, handwritten, and scene text; formulas as notation; and tables, layout, and reading order as structure. Key information extraction, question answering, translation, and retrieval are downstream tasks.

Recovering a table's cells, rows, and merged cells, or

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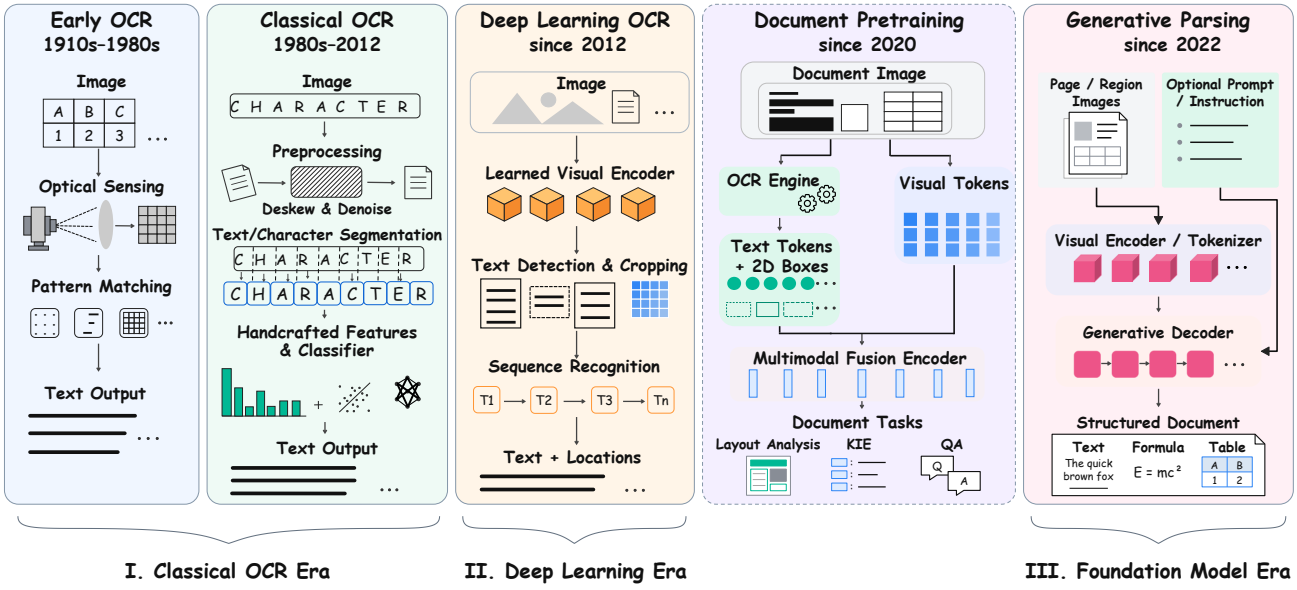

Fig. 2. **OCR and related document modeling across five historical tracks and three eras.** Columns show representative pipelines; braces mark three eras: **I. Classical OCR Era**, **II. Deep Learning Era**, and **III. Foundation Model Era**. The dashed **Document Pretraining** box provides related background: these models combine upstream OCR text, layout, and visual features for document understanding. Generative parsing reconstructs text, formulas, and tables within a unified structured document, using formats such as Markdown, HTML, LaTeX, or JSON. Dates indicate approximate emergence or historical prominence; the tracks overlap in time and several remain active in contemporary systems.

the values a chart encodes through marks and axes, reconstructs the source document; computing a statistic from those values or explaining a trend goes beyond it. Document understanding assigns semantic fields [12], answers questions [13], or draws conclusions about a document [14], [15]. We discuss it where it has shaped OCR architectures, training, or evaluation, and models that also perform understanding are examined through their reconstruction capabilities.

From specialist models to unified models: Fig. 2 places OCR and related document modeling in three eras. Early and Classical OCR matched optically sensed patterns against templates, and later used handcrafted features, statistical classifiers, and early neural networks [1], [16], [17], [18]. The Deep Learning Era replaced handcrafted features with learned ones and turned recognition into sequence prediction, so CRNN [19] no longer needed character segmentation; detection, layout, tables, and formulas kept dedicated models with their own data and outputs [5], [20], [21], [22], and a page was reconstructed by chaining them, as in toolkits such as PaddleOCR [23]. Unification began within this era. Mask TextSpotter [24] trained detection and recognition as one network, SPTS [25] generated locations and transcriptions as one sequence, and OmniParser [26] extended that sequence to key information extraction and table recognition, without relying on a pretrained LLM. In the same years, document pretraining models LayoutLM, DocFormer, and UDOP [27], [28], [29] fused text from an OCR engine with layout and visual features to model document structure. Their transcription came from that engine, and generative parsing set out to remove this dependence by decoding content and structure directly from the image, with compact encoder–decoders in Donut [7] for document understanding and Nougat [30] for full-page scientific transcription. GOT-OCR [8] and HunyuanOCR [31] built on pretrained LLM decoders to support several tasks and output formats. The unit of output became the page, and the structure that pipelines assembled by postprocessing is now generated, for the whole page at once or region by region after layout analysis. Specialist models and modular pipelines remain in use where accuracy, cost, or control favor them; Section 3 compares the five coexisting families.

Why does OCR need a new survey? Mori et al. [2] reviewed the historical development of OCR. Ye and Doermann [4], Zhu et al. [33], Long et al. [34], and Chen et al. [9] surveyed scene text detection and recognition across different stages of its development. Liu et al. [35] provided a broader review of intelligent document analysis and recognition. Recent surveys examine OCR with large models [36], document intelligence and parsing [106], [107], [108], text-centric image understanding [109], visual text processing and manipulation [37], [110], or general vision–language models that include OCR among many capabilities [38]. They adopt different boundaries for OCR.

In the Foundation Model Era, a single system can perform transcription, structural recovery, and downstream understanding, while specialist models, unified models, and hybrid workflows continue to coexist. Three questions follow: what belongs within OCR, how specialist and unified approaches are connected, and how their performance should be compared. This survey is written to answer them; Fig. 3 outlines its structure.

Contributions: (i) We systematically trace the development of OCR from specialist models toward unified models, focusing on the Foundation Model Era. (ii) We define OCR through faithful transcription and

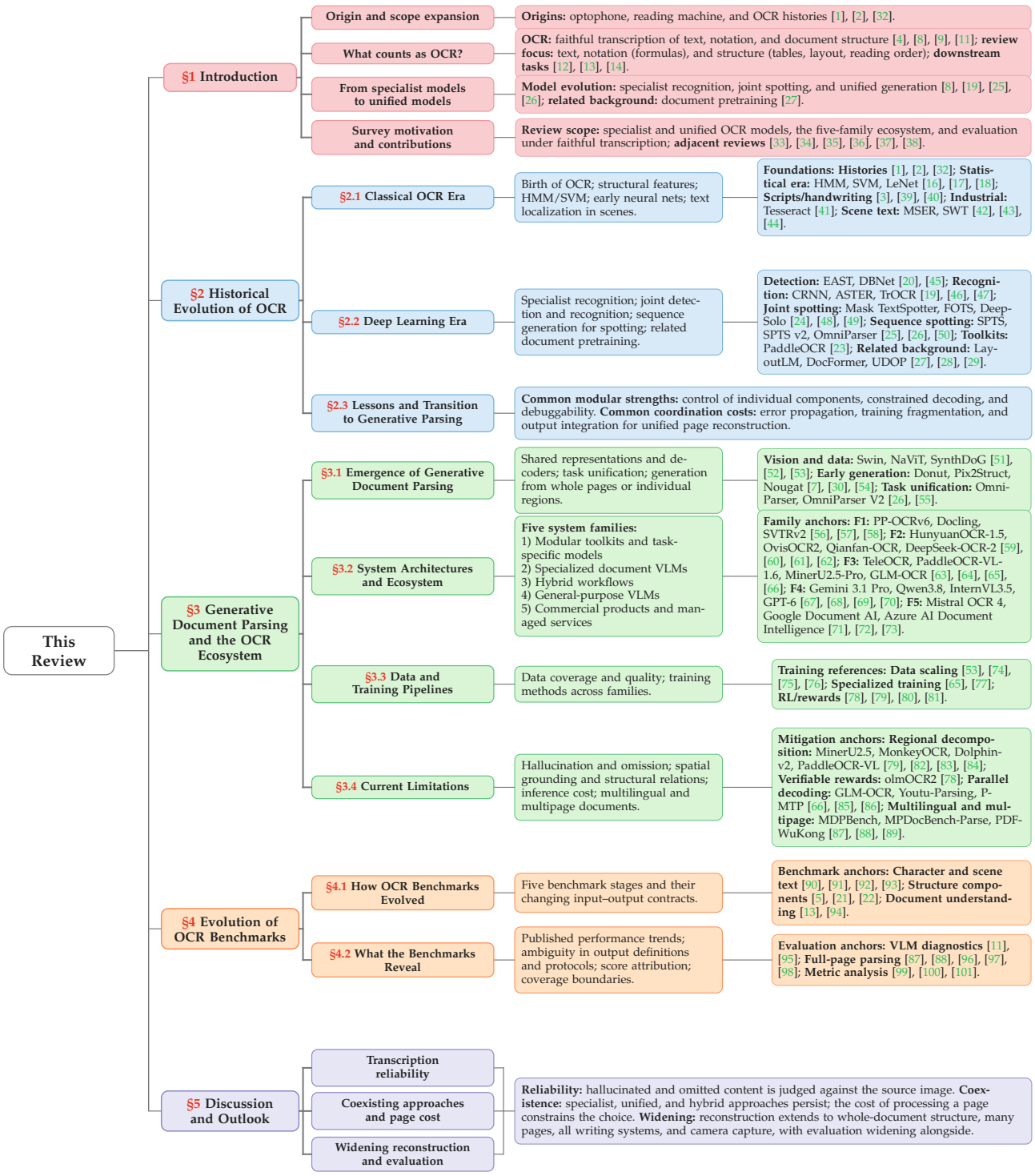

Fig. 3. **Review framework and evidence map.** The figure connects OCR's historical evolution and five contemporary system families with benchmark evidence, current limitations, and the three expectations of the outlook. Citations identify representative references.

compare the architectures and training of five coexisting system families. (iii) We summarize representative datasets and benchmarks, compare published results, and identify limitations of current evaluation. (iv) We discuss key challenges and future directions in transcription reliability, structural recovery, and processing efficiency.

This survey aims to help newcomers understand OCR and provide experienced researchers with a reference for recent advances and open problems.

2 Historical Evolution of OCR

This section reviews the classical and deep learning methods that underpin current OCR, emphasizing the move from specialist components to joint detection and recognition. Dedicated surveys cover these periods in detail [2], [4], [9], [33], [34], [35]. We focus on the technical changes relevant to later unified models, and discuss document pretraining as related background in document understanding. Figs. 2 and 4 place these developments in their historical context.

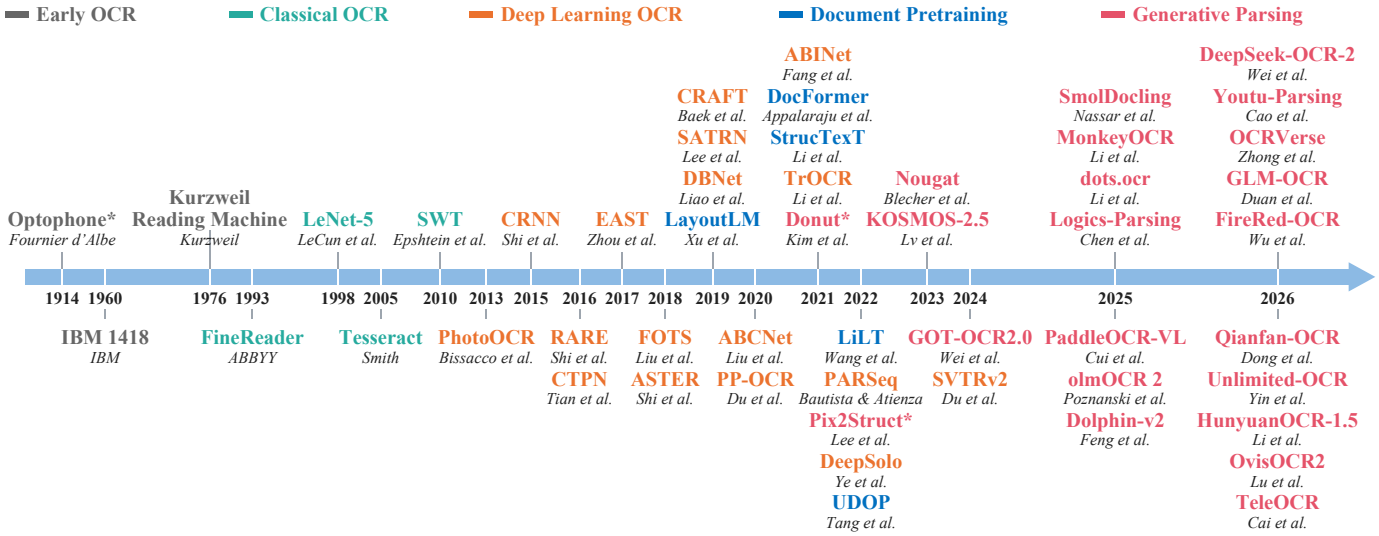

Fig. 4. Representative milestones in OCR and related document modeling, following the five tracks in Fig. 2. Years mark publications, public announcements, or system releases; horizontal spacing is schematic. Asterisks identify precursors: the optophone as a reading aid, and Donut and Pix2Struct in document understanding and image-to-sequence pretraining. Document Pretraining is included as background in document understanding. Section 2 covers classical and deep learning methods; Section 3 examines unified models and generative parsing.

2.1 Classical OCR Era

Machine reading predates electronic computing: Fournier d'Albe's *type-reading optophone* (1914) represented printed characters as tones for a listener to interpret, and Tauschek's *Reading Machine* (1929) matched characters against stored templates using a photodetector. During the 1950s and 1960s, commercial systems handled business document processing, bank check clearing, and postal sorting by constraining input to standardized fonts (the magnetic ink font E-13B, OCR-A, and OCR-B) and controlled imaging to produce digital character codes [1], [2].

As applications demanded broader font coverage, the field turned to statistical methods. HMMs [16] modeled character sequences, SVMs [17] classified segmented characters, and LeNet [18] demonstrated on handwritten digits that recognition features could be learned directly from pixel data. This era produced two defining systems: the commercial ABBYY FineReader (1993) [111] and Tesseract [41] (developed at HP Labs, released as open source in 2005, developed by Google in 2006–2017), both accurate on clean document scans. Recognition targets also broadened beyond machine print to handwriting [3] and non-Latin scripts [40], supported by corpora such as CASIA-HWDB [39].

Digital cameras then broke the paradigm's founding constraint: text appeared in scenes with uncontrolled fonts, backgrounds, and lighting. MSER [42] located character candidates as stable regions of high contrast, SWT [43] exploited the approximately constant stroke width of text, and the text spotter of Neumann and Matas [44] connected text localization with recognition in natural images. PhotoOCR [112] combined handcrafted features with a deep neural character classifier, marking a transition toward deep learning. Complex backgrounds and diverse typography challenged handcrafted features [4]. Advances in image classification [113], [114] highlighted the value of learned features.

2.2 Deep Learning Era

Deep learning replaced handcrafted features with learned representations and improved robustness to imaging conditions. Modular pipelines continued to use dedicated detection and recognition components, while joint models learned the two tasks together. In detection, CTPN [115] linked local text proposals; regression methods such as EAST [45] handled text in different orientations; later segmentation methods such as PSENet [116] and CRAFT [117] admitted arbitrary shapes; and DBNet [20] made binarization differentiable and became widely adopted in industrial deployments. End-to-end spotters such as FOTS [48] and Mask TextSpotter [24] combined detection and recognition in a single trainable network. Mask TextSpotter used character segmentation to recognize text within detected regions, supporting irregular shapes. ABCNet [118] used Bezier curves to describe curved text, while DeepSolo [49] used point queries to predict text geometry and characters jointly.

Sequence generation provided another way to unite localization and recognition. SPTS [25] represented each instance by a point and its transcription, generating both as one token sequence. SPTS v2 [50] first generated instance points and then recognized instances in parallel with shared decoder parameters, reducing sequential decoding cost. UNITS [119] used prompts to select geometric representations, allowing one text spotter to generate points, boxes, or polygons together with text. OmniParser [26] extended this sequence representation beyond spotting to key information extraction and table recognition (§3.1). These methods unified text spotting through a common output representation, with a trade-off between flexible sequence generation and decoding cost.

In recognition, CRNN [19] established the CNN–BiLSTM–CTC recipe. RARE [120] and ASTER [46] handle perspective distortion and text curvature through a learnable thin-plate-spline rectifier followed by sequence recognition with attention. Joint training on image–transcription pairs allows recognition supervision to guide geometric normalization without extra rectification annotations. ABINet [121] added an iterative language model branch; TrOCR [47] brought pretrained vision and language Transformers to recognition of cropped text; and SVTR [122] subsequently replaced the conventional CNN–RNN recognition backbone with a single visual Transformer. TrOCR's encoder–decoder design also anticipated the generative architectures of Section 3. Controlled studies in scene text recognition show that the scale and curation of training data can be as influential as decoder design [74], [75]. Further work on text recognition with deep learning addresses decoding efficiency, long text, complex typography, and training data: CPPD and IPAD study efficient parallel or iterative decoding [123], [124]; SMTR addresses text instances longer than those seen during training [125]; LMIM develops self-supervision informed by linguistics [126]; UnionST improves the synthesis of rendered text through corpus, font, and layout diversification and self-evolution labeling [127]; and WATERec combines support for inputs of arbitrary shapes with targeted synthetic data for artistic typography [128]. SCUT-EPT similarly exposes long text, erasures, supplements, switching, and noisy backgrounds [129]. Open-source toolkits (EasyOCR [130], docTR [131], MMOCR [132], and PaddleOCR [23]) assembled these components into deployable multilingual pipelines for detection and recognition; PaddleOCR further extended coverage to layout analysis and table and formula parsing.

Alongside OCR, document pretraining models (LayoutLM [27] and its successors [133], [134], DocFormer [28], and UDOP [29]) learned joint representations of OCR text, layout coordinates, and visual features for document understanding. They use transcriptions supplied by an upstream OCR engine. They are relevant here because they developed ways to combine text, layout, and vision and share representations across document tasks. Their downstream predictions also depend on the quality of the OCR input.

2.3 Lessons and Transition to Generative Parsing

For the recognition and reconstruction tasks in these lineages, faithful transcription remained the common goal as the visual scope expanded from isolated characters to scene images and structured document pages.

The two preceding eras transformed OCR from a constrained technology for specific domains into a broadly capable one: detection of text with arbitrary shapes, multilingual recognition, deployable table and formula parsing, and a mature open-source ecosystem. Modularity remains a genuine engineering asset: each component is independently testable, debuggable, and upgradable, and these pipelines remain in industrial use where reliability and cost control dominate.

Conventional modular pipelines face three common coordination problems. Their extent depends on representation sharing, joint training, and feedback across modules [23], [34]. (1) *Error propagation:* in predominantly feed-forward pipelines, a missed detection or incorrect layout label can affect subsequent recognition and document assembly. Joint training connects detection and recognition more closely [48], [49], but global correction across independently developed components is difficult [7], [8]. (2) *Training fragmentation:* detection, recognition, layout, table, and formula components are typically trained with different annotation schemas. Shared encoders, transfer learning, and multitask training help reuse data and representations [29], but differences in labels and training objectives make joint optimization across complete pages harder. (3) *Output integration:* standardized page schemas and common engineering frameworks help combine component outputs. Forming a coherent document requires aligning regions, recognized text, reading order, hierarchy, and specialized markup, usually through interface logic and postprocessing [8], [30].

These problems are particularly apparent in full-page reconstruction, where outputs must form a coherent document. Joint text spotters already reduce handoffs between detection and recognition. Generative document models extend integration to multiple content types and structural relations, using shared representations and structured outputs. Whole-page generation and layout-guided regional generation organize this integration differently (Section 3).

3 Generative Document Parsing and the OCR Ecosystem

Foundation models broaden the representations and capabilities that OCR tasks can share. Earlier end-to-end spotters jointly learned detection and recognition; document models extend sharing across text, notation, layout, and structured outputs. We examine three aspects of this unification: which tasks share a model, how their outputs represent content and structure, and how inference coordinates the page and its regions. A shared model may operate in several stages, as in layout prediction followed by regional recognition. We assess these choices through transcription fidelity, spatial and structural recovery, and inference cost, accounting for differences in architecture, training data, and decoding strategy.

This section traces how generative document parsing emerged (§3.1), organizes the current ecosystem into five coexisting system families and analyzes their architectural differences (§3.2), examines the data and training pipelines that further differentiate them (§3.3), and characterizes the limitations that define the current frontier (§3.4). The Generative Parsing track of Fig. 4 charts representative systems from Donut to recent models including OCRVerse [10], Unlimited-OCR [135], and TeleOCR [63].

3.1 Emergence of Generative Document Parsing

Generative document parsing drew on three advances outside OCR. Self-supervised pretraining at scale [76], [136] showed that a single model, trained on massive data, can learn transferable representations across tasks; Vision Transformers [137] brought this paradigm to images; and large language models tuned to follow instructions [138], [139] demonstrated that Transformer decoders can produce long,

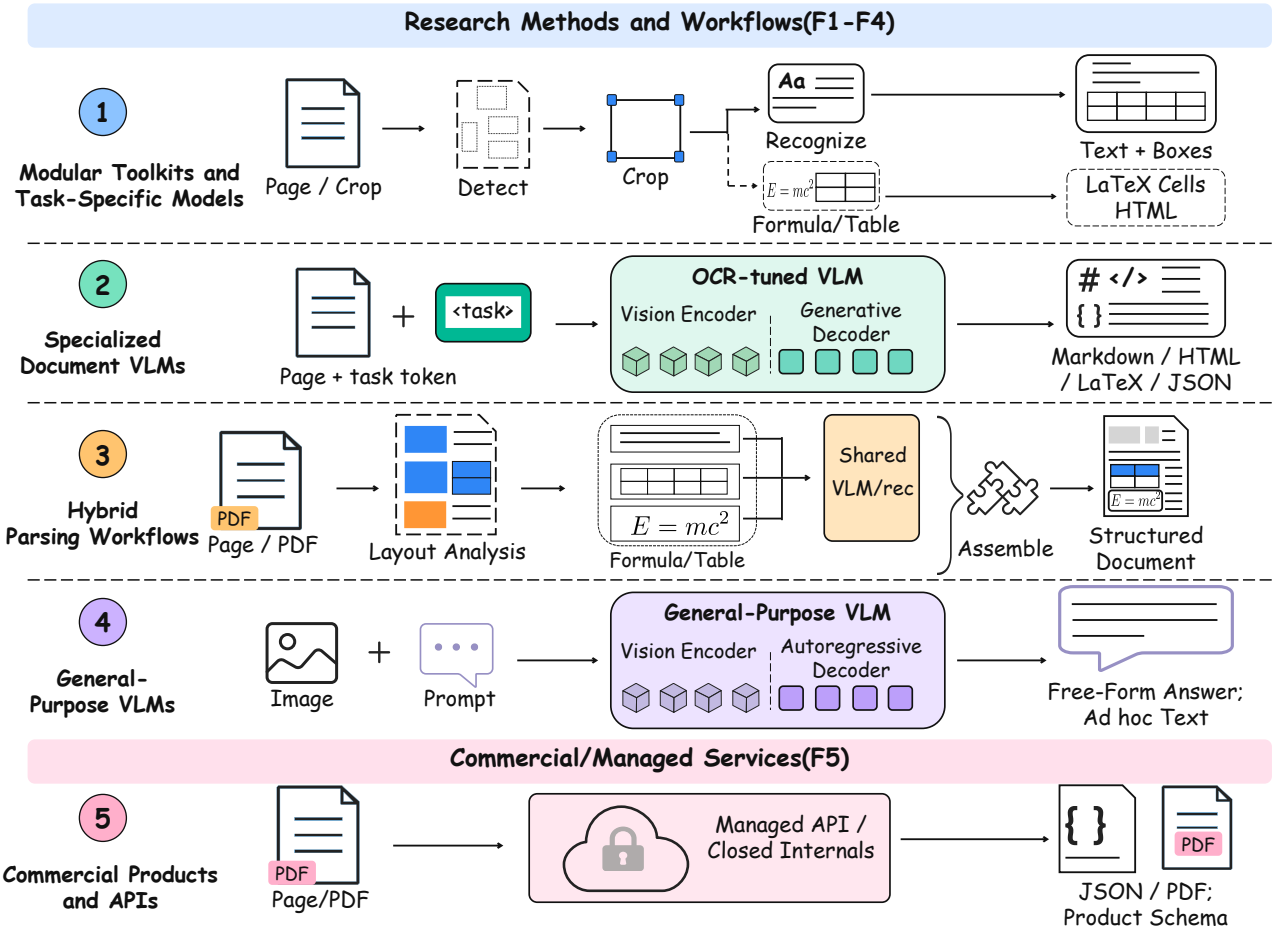

Fig. 5. Five system families in the OCR ecosystem of the Foundation Model Era, showing how each converts visual input into structured output. Each flow represents a family: **modular toolkits and task-specific models** combine task-specific prediction with modular processing, **specialized document VLMs** perform full-page generation with a model trained for documents, **hybrid parsing workflows** scaffold generation with explicit layout analysis and VLM recognition of individual regions, **general-purpose VLMs** expose transcription through open prompting, and **commercial products and managed services** expose closed stacks through product interfaces or managed endpoints. Table 1 lists representative systems.

structured text spanning thousands of tokens. These advances were general-purpose, but together enabled a single OCR model that reads a document image and generates its structured content end to end.

Three developments adapted this general capability to the demands of document OCR. First, vision encoders reached the resolution that dense document text demands: Swin Transformer [51] keeps attention cost tractable at page scale through hierarchical windowing, NaViT [52] processes images at native resolution and aspect ratio, and tiling schemes split a page into subimages so that encoders with a fixed input resolution, such as SigLIP [140], can be reused. Earlier encoders designed for scene understanding operated at 224–384 pixels [76], [137] and could miss small characters or fine table lines. Monkey [141] reused a pretrained encoder on local crops with separate low-rank adapters and a shared resampler, retaining a global image view. TextMonkey [142] introduced shifted-window attention for communication across windows and resampling guided by similarity to reduce redundant visual tokens. Second, the vision–language model (VLM) architecture emerged as a general framework for multimodal tasks [143], [144]: a vision encoder converts the image into visual tokens, a learned projector aligns them with the embedding space of a pretrained large language model, and the LLM generates text conditioned on them. We use *VLM* broadly for models that connect visual representations with language generation. The models discussed here include compact encoder–decoder precursors, such as Donut and Nougat, and recent systems built on pretrained LLM decoders. This architecture gave OCR a visual frontend with high capacity and a unified interface for structured generation. Third, training data for *whole pages* with structural annotations became available at scale. Rendering markup into images and automatically annotating real documents were established techniques [5], [6], [21], but they had supplied only markup for cropped regions or layout boxes for individual modules. Whole pages can be paired with transcription or markup sequences through synthetic document generation and alignment of existing PDFs with structured sources [30], [53], or through existing OCR pipelines with quality filtering [8], [78], supporting training on whole pages at scale.

Early models that map images to sequences without a separate OCR engine, such as Donut [7] and Pix2Struct [54], provided important technical foundations. Donut used an

encoder–decoder architecture with a decoder initialized from multilingual BART for task-specific document understanding. Pix2Struct used HTML generation from screenshots for pretraining across visual and linguistic tasks. Nougat [30] built on Donut's architecture to transcribe complete pages of scientific documents. Its target sequence represents both recognized content and document structure.

A complementary line unified prediction across document tasks through explicit spatial representations. OmniParser [26] used structure points to organize text spotting, key information extraction, and table recognition, followed by separate contour and text decoders. OmniParser V2 [55] added layout analysis and introduced a shared decoder with separate feedforward branches for structure, localization, and recognition. It also used its structured-points-of-thought representation to fine-tune multimodal language models. The transcription, localization, and structural reconstruction components fall within this survey's scope; semantic field extraction remains a downstream task. These methods show that task unification can precede the use of a large language model, and that explicit geometry can coexist with sequence generation.

For complete document reconstruction, whole-page generation places content and structure in a common decoding context. Dense pages, however, may require high input resolution and long output sequences. Layout-guided regional generation [82], [83], [84] addresses these costs by identifying regions and decoding their contents separately, retaining an explicit document assembly stage. These two routes draw on shared generative models and training methods and continue to develop in parallel.

3.2 System Architectures and Ecosystem

Recent benchmarks and technical reports have begun categorizing today's coexisting OCR systems into coarse groups such as pipelines, end-to-end systems, and VLMs [61], [96], [98]. A classical detection–recognition pipeline and a hybrid workflow that pairs a layout detector with generative recognizers are both pipelines, yet differ fundamentally in how they produce output; a specialized document VLM and a general-purpose VLM are both "end-to-end," yet differ in training objective, output consistency, and deployment constraints. We therefore refine the existing groupings into five families by asking two questions of each system. The first is how recognition and structure are integrated: through task-specific predictions produced within one model or combined by a toolkit or application (Family 1, including joint text spotters and specialized table or formula recognizers); at the region level, where an explicit layout stage selects regions for recognition and subsequent assembly (Family 3); or through structured generation over the whole page (Families 2 and 4). The second question separates the two families that generate whole pages by training objective: *specialized document VLMs* trained specifically for document OCR form Family 2, while *general-purpose VLMs* with OCR as one capability among many form Family 4. Commercial products and managed services often disclose too little architectural or training detail for reliable placement on either dimension. We therefore treat their delivery model as a separate category (Family 5). Table 1 marks compact encoder–decoder precursors with †: Donut primarily addressed document understanding, while Nougat adapted its architecture to transcription. The families describe OCR workflows; task sharing can occur within each family. Fig. 5 and Table 1 summarize the five families. The table selects historical landmarks, representative current architectures, and widely deployed platforms, and compares their primary tasks, typical outputs, parameter scale, and availability.

Family 1: Modular Toolkits and Task-Specific Models. Family 1 includes modular toolkits (Tesseract [41], PaddleOCR [23], [145], EasyOCR [130], docTR [131]) that chain detection, recognition, and postprocessing, as well as task-specific models (SVTR [122], UniMERNet [22], DocLayout-YOLO [146]) and joint text spotters. Such models solve defined recognition or localization tasks and can also serve as building blocks in Family 3. Their task-specific interfaces commonly support constrained or deterministic decoding: detection–recognition pipelines return text with spatial coordinates, table and formula modules emit structured markup, and layout models produce bounding boxes with category labels. These systems retain task-specific supervision and output formats and are widely used for stable workloads with clear requirements, where they offer predictable outputs and high throughput. Dedicated recognition remains active alongside OCR with foundation models. On WordArt-Bench, WATERec [128] substantially outperforms the VLM baselines evaluated under the same protocol, showing the continuing value of data and architectures tailored to the domain for challenging text distributions.

Family 2: Specialized Document VLMs. A specialized document VLM, designed or fine-tuned specifically for document OCR, maps page images to structured output (Markdown, HTML, LaTeX, or JSON), directly targeting the requirement for faithful parsing (Table 1). These systems commonly combine training focused on documents, visual encoding designed for OCR, and a predefined output schema. Structural relations such as heading hierarchy, table boundaries, and reading order are handled within the generative process, with designs ranging from implicit decoding [30] through chain-of-thought layout [61] to grounded multitask parsing [31]. GOT-OCR shares a language decoder across text and notation formats [8]; HunyuanOCR-1.5 uses task prompts to generate transcriptions, text coordinates, or document markup [59]. Unification therefore concerns both shared model parameters and support for different output requirements.

Family 3: Hybrid workflows. These systems bridge Families 1 and 2: an explicit layout stage, run by a dedicated detector (often a Family 1 component) or by the parsing model itself, segments the page into typed regions, and each region is decoded by a VLM recognizer before a postprocessor assembles the results into a structured document [65], [83], [84]. Working on bounded regions can shorten generation sequences, localize some errors, and make computational cost easier to budget. Reliability depends on layout recall and crop quality: missed or inaccurate regions may produce omissions that subsequent assembly cannot recover.

Hybrid workflows with VLM recognizers date from mid-2025. Dolphin first generated the page's element

TABLE 1
Representative systems in the OCR ecosystem across the five families (F1–F5). Year gives the first documented public appearance of each work, version, or product family. Tasks use a shared vocabulary. Outputs list representative content types or formats: Text includes answers and captions for downstream tasks, and Geometry covers boxes, polygons, and points. Params gives official nominal counts for the named model or family (the core VLM in F3); "–" marks undisclosed or non-unique values. † marks a compact encoder–decoder precursor; Donut is included for its role in document understanding and Nougat in transcription. Access icons link to official code, model hubs, papers, or services. Full inventory at https://github.com/zhihengli-casia/Awesome-OCR-in-the-Foundation-Model-Era.

Model / System	Year	Primary task	Typical output	Params	Access
F1: Modular toolkits and task-specific models					
Modular toolkits					
PP-OCRv6 [56]	2026	OCR	Text, Geometry	1.5–34.5M	
PP-StructureV3 [145]	2025	Document parsing	Markdown, JSON	–	
Docling [57]	2024	Document parsing	Markdown, JSON	–	
docTR [131]	2021	OCR	Text, Geometry	–	
EasyOCR [130]	2020	OCR	Text, Geometry	–	
Tesseract [41]	2005	OCR	Text, Geometry	–	
Task-specific models					
OmniParser [26]	2024	Spotting, KIE, table recognition	Text, Geometry, Markup	–	
SVTRv2 [58]	2024	Text recognition	Text	5.1/11.3/19.8M	
DocLayout-YOLO [146]	2024	Layout analysis	Geometry	19.5M	
UniMERNet [22]	2024	Formula recognition	LaTeX	107/202/325M	
DeepSolo [49]	2022	Text spotting	Text, Geometry	–	
DBNet++ [147]	2022	Text detection	Geometry	–	
TrOCR [47]	2021	Text recognition	Text	62/334/558M	
ASTER [46]	2018	Text recognition	Text	–	
CRNN [19]	2015	Text recognition	Text	8.3M	
F2: Specialized Document VLMs					
HunyuanOCR-1.5 [59]	2026	Document parsing	Text, Markdown, Geometry	1B	
OvisOCR2 [60]	2026	Document parsing	Markdown	0.8B	
Unlimited-OCR [135]	2026	Document parsing	Text, Geometry	3B (0.5B act.)	
Qianfan-OCR [61]	2026	Document parsing	Markdown, HTML, JSON	4B	
dots.mocr [148]	2026	Document parsing	Markdown, JSON, SVG	3B	
FireRed-OCR [149]	2026	Document parsing	Markdown	2B	
DeepSeek-OCR-2 [62]	2026	Document parsing	Markdown	3B (0.5B act.)	
olmOCR 2 [78]	2025	Document parsing	Text, Markdown	7B	
Granite-Docling [150]	2025	Document parsing	DocTags	258M	
GOT-OCR 2.0 [8]	2024	OCR	Text, Markup	580M	
Nougat† [30]	2023	Document parsing	Markdown	250/350M	
Donut† [7]	2021	Document understanding	Text, JSON	202M	
F3: Hybrid workflows					
TeleOCR [63]	2026	Document parsing	Markdown, JSON	1.2B	
MonkeyOCRv2-Parsing [77]	2026	Document parsing	Markdown, JSON	0.6/0.7B	
PaddleOCR-VL-1.6 [64]	2026	Document parsing	Markdown, JSON	0.9B	
MinerU 2.5-Pro [65]	2026	Document parsing	Markdown, JSON	1.2B	
GLM-OCR [66]	2026	Document parsing	Markdown, JSON	0.9B	
Youtu-Parsing [85]	2026	Document parsing	Text, LaTeX, HTML, Geometry	2.5B	
Dolphin-v2 [84]	2025	Document parsing	Markdown, JSON	3B	
MonkeyOCR [83]	2025	Document parsing	Markdown, JSON	3B	
F4: General-purpose VLMs					
GPT-6 [70]	2026	Multimodal understanding	Text, Markup	–	
Qwen3.8 [68]	2026	Multimodal understanding	Text, Markup	27B	
Gemini 3.1 Pro [67]	2026	Multimodal understanding	Text, Markup	–	
InternVL3.5 [69]	2025	Multimodal understanding	Text, Markup	1.1–241B	
F5: Commercial Products and Managed Services					
Mistral OCR 4 [71]	2026	Document parsing	Markdown, JSON	–	
Google Document AI [72]	2020	Document parsing	JSON	–	
Azure AI Doc Intelligence [73]	2019	Document parsing	JSON	–	
Amazon Textract [151]	2018	Document parsing	JSON	–	
Mathpix [152]	2017	Document parsing	Markdown	–	
ABBYY FineReader [111]	1993	Document parsing	Text, PDF	–	

sequence with a compact encoder–decoder and then parsed the elements in parallel with element-specific prompts [153]. MonkeyOCR (June 2025) is the earliest system in Table 1 to build such a workflow on a pretrained LLM decoder: a dedicated detector predicts the block structure, a 3B VLM recognizes each block, and a relation stage orders the blocks [83]. dots.ocr (July 2025) took the whole-page route, predicting layout and content with one VLM in a single pass [154]. Subsequent releases explored smaller recognizers and different degrees of sharing across stages: MinerU2.5 (September 2025) used a 1.2B VLM for layout at reduced resolution and for recognition from crops at native resolution [82]; PaddleOCR-VL-1.5 (January 2026) kept the two-stage form with a 0.9B VLM [79]; and Dolphin-v2 fine-tuned one VLM for both layout analysis and content parsing, with regional recognition for digital documents and whole-page parsing for photographed ones [84]. Releases in 2026 kept this structure and moved effort to training and decoding: MinerU2.5-Pro added fine-tuning on difficult samples and PaddleOCR-VL-1.6 progressive post-training [64], [65]; MonkeyOCRv2 replaced the reused vision encoder with one pretrained from scratch on document images [77] (§3.3); GLM-OCR and Youtu-Parsing cut sequential decoding steps with multi-token and parallel decoding [66], [85] (§3.4); and TeleOCR extended the workflow to camera-captured documents [63]. Recognizers in this family now range from

0.6B to 3B parameters, with most 2026 releases below 1.5B.

Family 4: General-purpose VLMs. General-purpose VLMs (GPT-6 [70], Gemini 3.1 Pro [67], Qwen3.8 [68], InternVL3.5 [69]) are trained for broad understanding of visual and linguistic content, with OCR as one capability among many. These models support transcription through prompts, alongside applications such as question answering about images and semantic field extraction. The share of documents in their training mixtures is generally undisclosed. On OmniDocBench v1.6_full, the general-purpose VLMs on the official leaderboard span 83.76 to 93.70 Overall, from InternVL3.5-241B to Ovis2.6-30B-A3B, and the strongest of them falls within the 93.26–96.91 range of the specialized and hybrid systems in Table 6 [155].

Family 5: Commercial Products and Managed Services. Commercial products and managed services (ABBYY, Mathpix, TextIn, Mistral OCR 4 [71], Google Document AI [72], Azure [73]) typically emphasize operational reliability, format control, integration support, and enterprise deployment features. Where documentation describes the processing, it maps onto the preceding families: Document AI and Azure AI Document Intelligence expose separate reading, layout, and form or table processors in the manner of Family 1 toolkits, while Mistral OCR 4 serves one document model that returns Markdown and JSON. Internal architectures and training data remain undisclosed, so the family is defined by its delivery model and described through its documented interfaces (Table 1).

3.3 Data and Training Pipelines

The capabilities of each system family depend on both its architecture and the data and training signals from which those capabilities are learned. Fig. 6 compares these two dimensions across the four trainable families. Local transcription labels supervise recognition, while page reconstruction also requires reading order, hierarchy, and relations between elements.

Data sources: Data quantity and quality are established performance variables in scene text recognition [74], [75], and the four trainable families draw on progressively broader data (Fig. 6, left). Family 1 trains each module on *task-specific data*: scene text images, word and line crops, pages with layout annotations, table crops, and formula crops. Public corpora range from roughly 10^5 annotated pages or formulas to nearly 10^7 synthetic word images per task [5], [6], [21], [156]. Family 2 assembles *data centered on documents*, with reported training mixtures spanning several million to tens of millions of page and region image–text pairs [7], [8], [30], [157]. These data combine full pages with region crops through synthetic rendering [53], engine bootstrapping, deduplication using CLIP [76], quality filtering, and curriculum schedules. Family 3 combines *layout annotations* (pages with layout labels and bounding boxes) with *cropped regions* paired with transcription targets. Representative systems report several million to several tens of millions of mixed training samples, while their subsets for layout supervision range from tens of thousands to a few million pages [79], [82], [83]. Family 4 trains on *broad multimodal corpora* spanning photographs, charts, conversations, code, and documents, where document understanding is one of many target capabilities. Open technical reports describe hundreds of billions to several trillion pretraining tokens, followed by roughly one to tens of millions of supervised multimodal examples [158], [159], [160]; the proportions of documents are generally undisclosed.

Training pipelines: Training methods also diverge across families (Fig. 6, right). Family 1 uses *task-specific supervision*: CTC or token cross-entropy for recognizers [19], [122], and detection or layout losses for other modules [20], [21]. Components may be trained separately, while end-to-end spotters share features and jointly optimize localization and recognition [24], [25]. Family 2 is typically trained in two stages, *pretraining* for alignment between document images and text followed by *supervised fine-tuning* (SFT) on document tasks covering text fidelity, table structure, and formula syntax; a growing number of systems add a third stage of *reinforcement learning with verifiers* that exploits OCR's distinctive property of producing verifiable outputs. These objectives include rewards from executable unit tests for text presence and absence, reading order, tables, formulas, and abnormal repetition in olmOCR2 [78] and rewards that account for output format [81]. Their effects are bounded by the fidelity properties encoded in the reward. Family 3 uses either *separate models* or a *shared VLM* for layout analysis and recognition. With separate models, the layout detector is trained on annotated pages, while region parsers follow Family 2's training pipeline or are reused from existing models. Dolphin-v2 fine-tunes the same VLM for both layout analysis and content parsing [84]. PaddleOCR-VL-1.5 also uses GRPO to reduce inconsistency in output style [79], [80]. Within Family 3, MonkeyOCRv2 pretrains standalone ViT-Small/Base encoders from scratch on document images, using text and pixel reconstruction decoders that share the visual tokens [77]. For parsing, it freezes the encoder and connects it to Qwen3-0.6B through an MLP projector. Family 4 follows a pipeline with four stages: *visual alignment* (connector warmup), *joint pretraining* on multimodal and text data, *SFT* via instruction tuning, and optional *post-training* with DPO, RLHF, or RLVR. In downstream document question answering, DocThinker [161] uses rule-based reinforcement learning with rewards for answer correctness, supporting-region localization, question reformulation, and output format.

3.4 Current Limitations

Families 2–4 generate output conditioned on the image. Learned linguistic and layout regularities should support recovery of the content in the source image, but when compression, limited resolution, or unfamiliar layouts weaken the visual evidence, generation may rely on learned priors, producing unsupported content or omitting difficult regions. Localization, reading order, and relations across regions add requirements beyond recognizing the text, and high input resolution with long structured outputs increases computation. Multilingual text and documents with many pages combine these difficulties.

Hallucination and omission: Hallucination and omission can arise when visual detail is lost, complex layouts are only partly covered, or decoding errors accumulate in long sequences [7], [8], [96]. Published evaluations

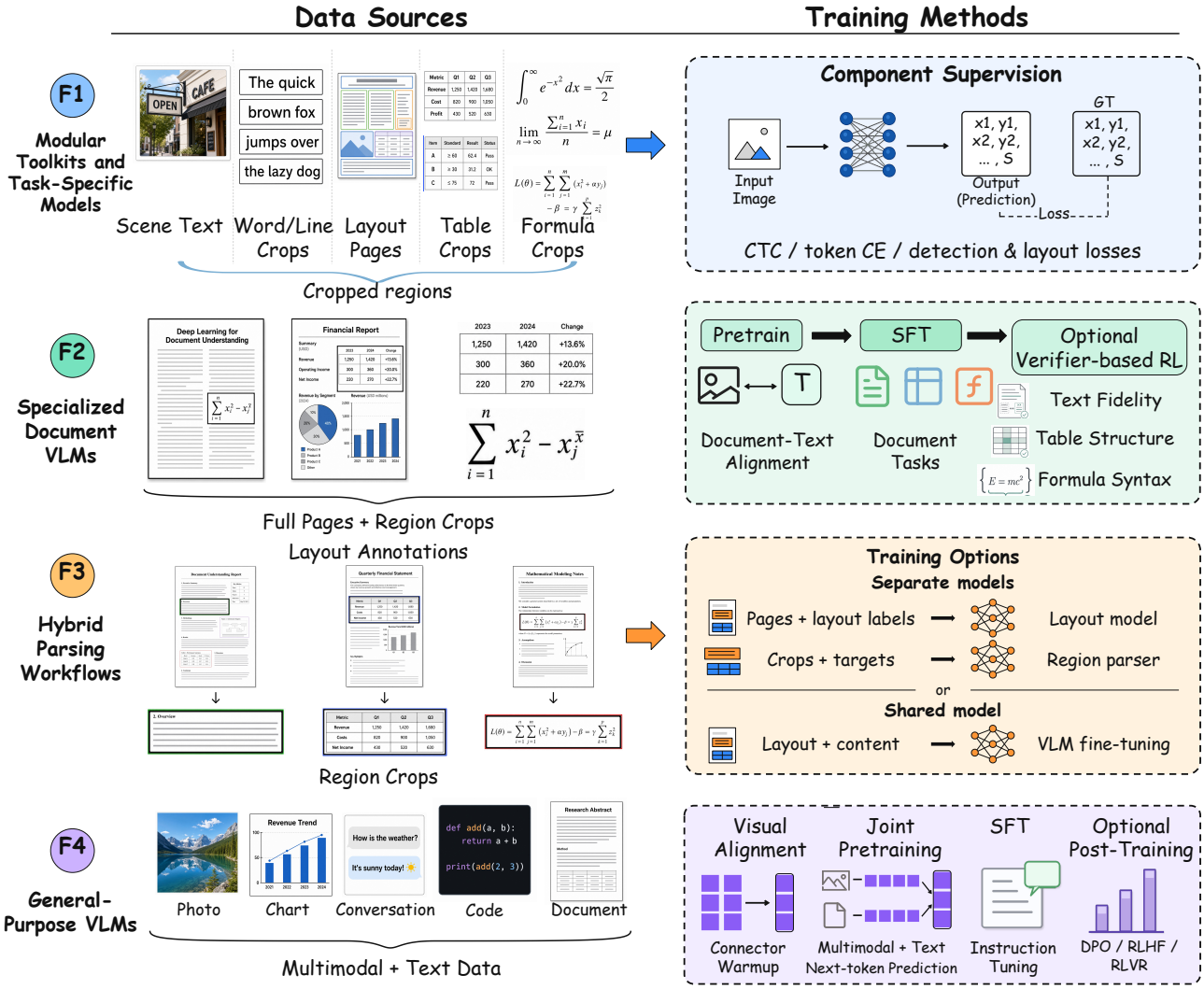

Fig. 6. Data sources (left) and training pipelines (right) for the first four system families. **Modular toolkits and task-specific models** use task-specific data and supervision; **specialized document VLMs** use pages and region crops for pretraining, supervised fine-tuning, and optional reinforcement learning with verifiers; **hybrid parsing workflows** train separate layout and recognition models or a shared VLM for both tasks; and **general-purpose VLMs** use broad multimodal corpora with alignment, joint pretraining, instruction tuning, and optional post-training.

report fabricated text and phantom table rows, as well as omissions, collapsed columns, and incorrect heading levels [96], [98]. MinerU2.5, MonkeyOCR, PaddleOCR-VL, and Dolphin-v2 use layout analysis and parsing of individual regions to shorten generation sequences and preserve local detail [79], [82], [83], [84]; MinerU2.5-Pro and olmOCR2 strengthen output fidelity through data verification and executable rewards [65], [78]. The olmOCR2 study reports that a data loader bug excluded blank pages from training and caused hallucinations on blank inputs, an error that benchmark scores did not register, since fixing it and retraining left them unchanged [78].

Spatial grounding and structural relations: Spatial grounding associates each output element, such as text, a formula, a table, or a layout block, with its region in the source image through a bounding box, coordinates, or a region identifier. It makes generated content verifiable against the image and supports reconstruction of reading order, hierarchy, and relations across regions. Its precision is bounded by the visual detail the encoder retains and by coordinate quantization in the output [31], [61]. Benchmark evaluations show that general-purpose VLMs remain limited in text localization, layout perception, and parsing of complex elements [95], and plain Markdown, HTML, or LaTeX output generally omits the correspondence between elements and image regions altogether [8], [30]. Hybrid workflows add explicit structure: MonkeyOCR separates structure detection, region recognition, and reading-order prediction through its Where–What–How formulation [83]; Dolphin-v2 uses absolute coordinates and layout anchors to improve crop precision; and PaddleOCR-VL predicts regions defined by multiple points jointly with reading order and grounded text spotting [79], [84].

Inference cost: Inference cost arises from visual input at high resolution and from long structured output: dense pages produce many visual tokens, and generating one token at a time introduces latency that grows with output length. GLM-OCR combines multi-token prediction with parallel region recognition [66], Youtu-Parsing introduces token parallelism and query parallelism over

shared page features [85], and P-MTP progressively increases lookahead depth and dynamically selects speculative lengths during inference [86]. Parallel processing of regions and multi-token decoding reduce sequential generation steps, while layout processing, scheduling, verification, and assembly add system overhead.

Multilingual text: Language coverage expanded in the earlier eras through dedicated datasets: CASIA-HWDB and IFN/ENIT for Chinese and Arabic handwriting [39], [162], the ICDAR 2019 MLT challenge for detection and recognition across ten languages in scene images [163], and XFUND for form understanding in seven languages [164]. Generative parsers inherit their coverage from the pretrained language model and from the documents added in fine-tuning (§3.3), and evaluations show where it falls short: the question-answering benchmark MTVQA reports low accuracy of leading VLMs on text-centric questions in nine languages [165], CC-OCR includes a multilingual reading track [166], and MDPBench reports diacritic omissions, character substitutions across writing systems, and errors in ordering text from right to left in document parsing [87].

Long and multipage documents: Most parsing benchmarks and training data use single pages (Table 2). A document with many pages requires continuing paragraphs and tables across page breaks, recovering a heading hierarchy over the whole document, and maintaining reading order across pages, while visual tokens and output length accumulate with page count. MPDocBench-Parse evaluates these requirements and reports weaknesses in merging content across pages and in recovering hierarchy [88]. Long-document understanding lies outside OCR under our definition but has already faced the same cost of many pages: DUDE and MMLongBench-Doc pose questions whose evidence is spread over many pages [167], [168], PDF-WuKong parses the document into text and image blocks and then selects evidence by sparse sampling conditioned on the query [89], and DocSeeker learns to locate evidence pages through distilled supervision and reinforcement learning [169]. These methods focus on evidence relevant to a question, while complete parsing requires content and structure across all pages, so output representations that carry reading order and hierarchy across page boundaries, and evaluation on complete documents, remain open.

4 Evolution of OCR Benchmarks

The system evolution of Section 3 changed what OCR benchmarks must measure. The required output has grown from a class label or a character string to a structured reconstruction of a whole page or document, with locations, reading order, tables, formulas, and markup returned together. Evaluation followed this growth in stages, each opened when systems outgrew the measure before it, and none closed by its successors. Section 4.1 traces the five stages; Section 4.2 reads published results as of October 2026 against them.

4.1 How OCR Benchmarks Evolved

Five stages trace this development (Fig. 7 and Table 2): the input expands from isolated characters to complete documents, and the output from class labels through region transcripts and component structures to full-document markup.

The first stage tested *controlled text recognition*. MNIST [18] reduced the problem to classifying single digits, while IAM [170] and CASIA-HWDB [39] extended evaluation to handwritten characters and complete text lines. These protocols receive the text already located and cropped; finding it defined the next stage.

The second stage brought OCR into natural scenes. The ICDAR Robust Reading competitions [91], [171] and later datasets such as Total-Text [92] and CTW1500 [93] established protocols for *scene text detection and end-to-end spotting*, and word-crop benchmarks IIIT5K [172], SVT [90], SVTP [173], and CUTE80 [174] isolated *scene text recognition*. Video text was probed through the ICDAR 2013 video challenge [171] and BOVText [175], outside the dominant lineage. The unit of judgment remained the word or region; HierText [176] grouped words into lines and paragraphs, while tables, formulas, and page-level structure stayed outside these protocols.

The third stage, *document structure components*, brought structure into the metric, one component at a time. PubTabNet [5] evaluated table structure as HTML with TEDS, FinTabNet [177] paired cell boxes with logical row–column structure and evaluated cell adjacency, and CROHME [178] and UniMER-Test [22] evaluated formula recognition under structural or LaTeX conventions. Layout benchmarks followed a different contract: PubLayNet [21] and DocLayNet [179] formulate the task as localization and classification of predefined page regions and report mAP within their respective label spaces. Each component is scored under its own definition of structure, with no shared protocol across components.

The fourth stage, *document understanding*, judges systems by their answers. DocVQA [13], ChartQA [94], and DUDE [167] pair document images with natural language questions, while FUNSD [180] and CORD [181] target key information extraction. Under the boundary in Section 1 these tasks are downstream applications, and a correct answer does not verify the transcription behind it; they belong in this lineage through their influence on VLM system design.

The fifth stage, *OCR evaluation in the Foundation Model Era*, returned to reconstruction and divides into two tracks. OCRBench [11] and OCRBench v2 [95] diagnose VLM abilities across transcription, localization, and understanding tasks. OmniDocBench [96], [155], Real5-OmniDocBench [97], PureDocBench [98], MDPBench [87], and MPDocBench-Parse [88] evaluate reconstruction of complete pages and multipage documents across content fidelity, reading order, and document structure, placing multiple system families on shared pages while retaining metrics for individual components. With this stage the unit of evaluation caught up with the unit of system output: the page, and most recently the document.

4.2 What the Benchmarks Reveal

4.2.1 Quantitative Performance Trends

Across these stages, published results show saturation moving outward: scene text first, structural components next,

TABLE 2
Representative benchmarks across five stages of evaluation relevant to OCR (newest first within each subgroup). Year gives the first public paper or dataset release, or the named version release. Input uses Page for a single page, Multipage for a document, and Mixed for heterogeneous inputs. Geometry denotes boxes or polygons. Main metrics follow official protocols; ED denotes edit distance, CDM evaluates formulas, and TEDS evaluates tables. Scale refers to the named release or split; K/M counts are rounded and test splits are marked. Access icons link to official repositories, dataset pages, or papers. Full list: project page.

Benchmark	Year	Input	Output	Main metrics	Scale	Access
❶ Controlled text recognition						
Character classification						
EMNIST [182]	2017	Crop	Labels	Accuracy	814K images	
SVHN [183]	2011	Crop	Labels	Accuracy	600K+ images	
MNIST [18]	1998	Crop	Labels	Accuracy	70K images	
Handwriting character, line, and word recognition						
CASIA-HWDB 1.x/2.x [39]	2011	Page, Crop	Text, Labels	Accuracy, CR/AR	3.9M chars, 5.1K pages	
RIMES 2011 (line) [184]	2011	Crop	Text	CER, WER	12,111 lines	
IAM [170]	2002	Crop	Text	CER, WER	13.4K lines	
IFN/ENIT [162]	2002	Crop	Text	Word acc.	26,459 images	
❷ Scene text						
Detection and spotting						
HierText [176]	2022	Scene	Masks, Hierarchy	PQ	11,639 images	
TextOCR [185]	2021	Scene	Text, Geometry	Det. F1, Word acc.	28,134 images	
ReCTS [186]	2019	Scene	Text, Geometry	Hmean, 1-NED	25K images	
LSVT [187]	2019	Scene	Text, Geometry	Hmean, 1-NED	450K images (400K weak)	
ArT [188]	2019	Scene	Text, Geometry	Hmean, 1-NED	10K images	
MLT 2019 [163]	2019	Scene	Text, Geometry, Script	Hmean, Script acc., 1-NED	20K images	
CTW1500 [93]	2017	Scene	Geometry	P/R/F1	1.5K images	
Total-Text [92]	2017	Scene	Geometry	P/R/F1	1.6K images	
MLT 2017 [189]	2017	Scene	Geometry, Script	Hmean, Script acc.	18K images	
COCO-Text [190]	2016	Scene	Geometry	P/R/F1	63K images	
ICDAR 2015 [91]	2015	Scene	Geometry	P/R/F1	1.5K images	
ICDAR 2013 [171]	2013	Scene	Geometry	P/R/F1	462 images	
MSRA-TD500 [191]	2012	Scene	Geometry	P/R/F1	500 images	
Cropped word recognition						
Union14M-Bench [75]	2023	Crop	Text	Word acc., WAICS	409K images (test)	
ICDAR 2015 (rec.) [91]	2015	Crop	Text	Word acc.	2,077 images (test)	
CUTE80 [174]	2014	Crop	Text	Word acc.	288 images (test)	
SVTP [173]	2013	Crop	Text	Word acc.	639 images (test)	
ICDAR 2013 (rec.) [171]	2013	Crop	Text	Word acc.	1,095 images (test)	
IIIT5K [172]	2012	Crop	Text	Word acc.	3,000 images (test)	
SVT [90], [192]	2010	Crop	Text	Word acc.	647 images (test)	
Video text spotting						
BOVText [175]	2021	Video	Text, Tracks	MOTA, MOTP, IDF1	2,021 videos, 1.75M frames	
❸ Document structure components						
Tables						
PubTables-1M [193]	2021	Page, Crop	Geometry, Structure	AP, GriTS	947.6K tables	
WTW [194]	2021	Scene, Page	Geometry, Structure	F1, TEDS	14,581 images	
FinTabNet [177]	2020	Crop	Geometry, Structure	Adjacency F1	112,887 tables	
PubTabNet [5]	2019	Crop	HTML	TEDS	568K tables	
SciTSR [195]	2019	Crop	Cell graph	Structure F1	15K tables	
TableBank [196]	2019	Page, Crop	HTML, Geometry	P/R/F1, BLEU	417K det., 145K struct.	
Formulas						
UniMER-Test [22]	2024	Crop	LaTeX	ExpRate, ED, CDM	23,757 images	
HME100K [197]	2022	Crop	LaTeX	ExpRate	100K images	
CROHME 2016 [178]	2016	Strokes	Symbols, Relations	ExpRate	1,147 expressions (test)	
Im2LaTeX-100K [6]	2016	Crop	LaTeX	BLEU, Image ED/match	100K images	
Layout element detection						
DocLayNet [179]	2022	Page	Geometry, Labels	COCO mAP	80,863 pages	
DocBank [198]	2020	Page	Token labels	F1	500K pages	
PubLayNet [21]	2019	Page	Geometry, Labels	COCO mAP	360K pages	
❹ Document understanding†						
Document and scene text VQA						
OCR-Reasoning [199]	2025	Mixed	Answer, Rationale	Accuracy, Reasoning score	1,069 QA	
DUDE [167]	2023	Multipage	Answer, Confidence	ANLS, ECE, AURC	41.5K QA	
ChartQA [94]	2022	Chart	Answer	Relaxed acc.	32.7K QA	
DocVQA [13]	2020	Page	Answer	ANLS	50K QA	
TextVQA [200]	2019	Scene	Answer	VQA acc.	45.3K QA	
Key information extraction						
XFUND [164]	2021	Page	Entities, Relations	Entity/Relation F1	1.4K forms	
FUNSD [180]	2019	Page	Entities, Relations	Entity/Relation F1	199 forms	
❺ Foundation model OCR evaluation						
VLM OCR ability diagnosis						
CC-OCR v2† [201]	2026	Mixed	Text, Structure, Geometry, Fields, Answer	Task scores, Mean	7,093 samples	
OCRBench v2† [95]	2024	Mixed	Task outputs	Task scores, Total	10K QA	
CC-OCR† [166]	2024	Mixed	Text, Structure, Geometry, Fields	Task scores, Mean	7,058 images	
MTVQA† [165]	2024	Mixed	Answer	Accuracy	28,607 QA	
OCRBench† [11]	2023	Mixed	Task outputs	Total score	1K QA	
Full-page parsing						
PureDocBench [98]	2026	Page	Markdown	Text ED, CDM, TEDS, Order ED	1,475 pages, 4,425 images	
ParseBench† [202]	2026	Page, Multipage	Markdown, HTML, Structured data, Geometry	GTRM, ChartDataPointMatch	2,078 pages	
OmniDocBench v1.6 [96], [155]	2026	Page	Markdown	Text ED, CDM, TEDS, Order ED	1,651 pages	
MDPBench [87]	2026	Page	Markdown	Text ED, CDM, TEDS, Overall	3,400 images	
Real5-OmniDocBench [97]	2026	Page	Markdown	Text ED, CDM, TEDS, Order ED	1,355 pages, 6,775 images	

† Stage ❹ is a transition through downstream document understanding; marked suites involve semantic extraction or reasoning beyond faithful transcription.

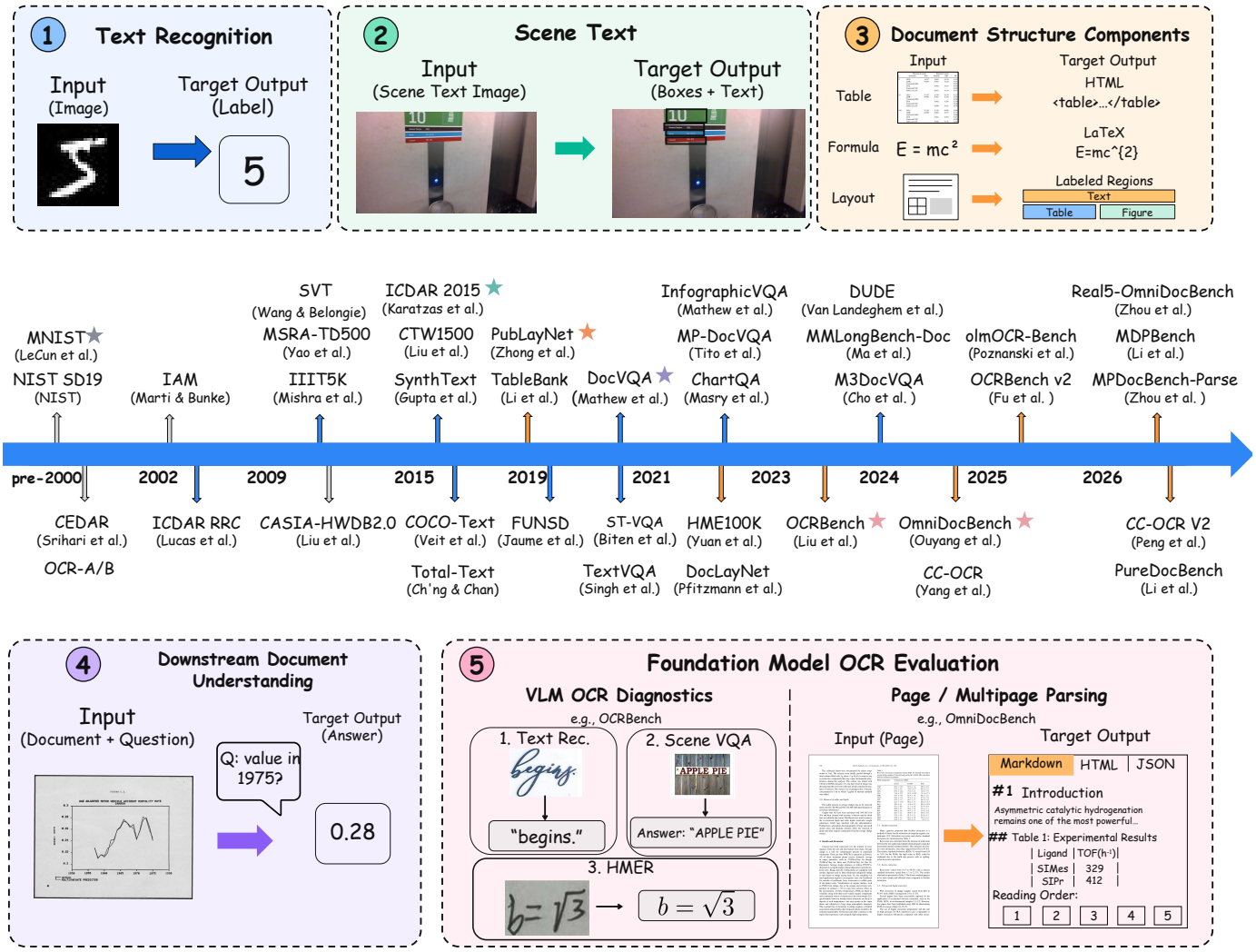

Fig. 7. Evolution of evaluation relevant to OCR across five stages, combining representative input–output contracts with a chronological timeline of benchmark datasets. ❶ *Text recognition*: controlled characters, words, and lines (MNIST, IAM, CASIA-HWDB). ❷ *Scene text*: detection and recognition in natural images (ICDAR, SVT, Total-Text, CTW1500). ❸ *Document structure components*: tables and formulas as structured markup, plus layouts as labeled spatial regions (PubTabNet, PubLayNet, DocLayNet, HME100K). ❹ *Document understanding*: question answering about documents (DocVQA, ChartQA, DUDE). ❺ *Foundation Model OCR Evaluation*, split into two tracks: VLM OCR ability diagnostics (OCRBench and OCRBench v2) and benchmarks for parsing full pages or multipage documents (OmniDocBench, Real5-OmniDocBench, MDPBench, MPDocBench-Parse, and PureDocBench). Document understanding and VQA form a transitional stage here and are discussed as downstream tasks that influence OCR model design and evaluation. These evaluation directions expand and coexist; VLM diagnostic suites may combine transcription and understanding tasks. Stars mark selected landmark benchmarks; Table 2 lists dataset details and metrics.

TABLE 3
Scene text detection Hmean/F1 (%).

Method	Date	IC15	Total	CTW
PSENet [116]	2018.06	85.7	80.9	82.2
TextSnake [203]	2018.07	82.6	78.4	75.6
PAN [204]	2019.08	82.9	85.0	83.7
DBNet [20]	2019.11	87.3	84.7	84.9
TextFuseNet [205]	2020.07	**92.1**	87.1	86.6
FCENet [206]	2021.04	86.2	85.8	85.5
TESTR [207]	2022.04	90.0	86.9	87.1
TPD-STR [208]	2025.02	89.4	**88.7**	**88.5**

TABLE 4
Scene text recognition accuracy (%) under MJ+ST, following the comparison compiled in SVTRv2 [58]. IC13/IC15 use the 857/1811-image subsets.

Model	Date	Params	IIIT5K	SVT	IC13	IC15	SVTP	CUTE
CRNN [19]	2015.07	8.3 M	82.9	81.6	91.1	69.4	70.0	65.5
ASTER [46]	2018.06	27.2 M	93.3	90.0	90.8	74.7	80.2	80.9
ABINet [121]	2021.03	36.7 M	96.2	93.5	**97.4**	86.0	89.3	89.2
SVTR-B [122]	2022.05	24.6 M	96.0	91.5	97.1	85.2	89.9	91.7
PARSeq [105]	2022.07	23.8 M	97.0	93.6	97.0	86.5	88.9	92.2
SVTRv2-B [58]	2024.11	19.8 M	**97.7**	**94.0**	97.3	**88.1**	**91.2**	**95.8**

and page-level text on OmniDocBench by 2026, while table structure within pages and physical capture conditions still separate systems.

Scene text. CTW1500 detection Hmean increased from 75.6 for TextSnake to 88.5 for TPD-STR; under the MJ+ST setting, IIIT5K recognition accuracy increased from 82.9 for

TABLE 5
End-to-end scene text spotting F1 (%). Dates refer to the first public papers. "None" is lexicon-free; "Full" uses all words in the test set.

		IC15	Total-Text		CTW1500	
Method	**Date**	**None**	**None**	**Full**	**None**	**Full**
FOTS [48]	2018	–	–	–	21.1	39.7
Mask TextSpotter v2 [209]	2019	–	65.3	77.4	–	–
TextDragon [210]	2019	–	48.8	74.8	39.7	72.4
ABCNet [118]	2020	–	64.2	75.7	45.2	74.1
ABCNet v2 [211]	2021	–	70.4	78.1	57.5	77.2
TESTR [207]	2022	65.3	73.3	83.9	56.0	81.5
DeepSolo [49]	2022	73.8	83.6	89.6	64.2	81.4
ESTextSpotter [212]	2023	–	80.8	87.1	64.9	83.9
DG-Bridge [213]	2024	–	83.3	88.3	**69.8**	83.9
TiCLS [214]	2026	**79.1**	**84.6**	**90.4**	66.2	**88.1**

TABLE 6
Selected OmniDocBench v1.6_full scores [155] as of October 2026. Overall/Text/Table/Fml. are % (↑); Text is $(1-\mathrm{Edit})\times 100$, Table is TEDS, Fml. is CDM, and RO is reading order Edit (↓).

System	**Date**	**Params**	**Overall**	**Text**	**Table**	**Fml.**	**RO**
Specialized Document VLMs							
OvisOCR2 [60]	2026.07	0.8 B	96.47	**97.35**	94.58	97.49	**0.112**
Unlimited-OCR [135]	2026.06	3 B	94.00	96.06	90.21	95.72	0.128
Qianfan-OCR [61]	2026.03	4 B	93.90	96.00	90.53	95.08	0.130
FireRed-OCR [149]	2026.02	2 B	93.26	96.30	88.04	95.44	0.131
Hybrid workflows							
TeleOCR [63]	2026.08	1.2 B	**96.91**	97.33	**96.82**	96.59	0.118
PaddleOCR-VL-1.6 [64]	2026.05	0.9 B	96.34	96.74	94.76	**97.53**	0.128
MinerU2.5-Pro-2605 [215]	2026.05	1.2 B	95.72	96.40	93.62	97.15	0.123
GLM-OCR [66]	2026.02	0.9 B	95.22	95.60	92.83	97.18	0.133

TABLE 7
Selected Real5-OmniDocBench leaderboard scores [216] (%,↑), as of October 2026, using the pinned OmniDocBench v1.5 protocol. S/W/SP/I/SK = scan/warp/screen/illum./skew; All averages the five condition scores; Worst is their minimum.

System	**All**	**S**	**W**	**SP**	**I**	**SK**	**Worst**
Modular toolkits and task-specific models							
PP-StructureV3 [145]	64.45	84.68	59.34	66.89	73.38	37.98	37.98
Specialized Document VLMs							
OvisOCR2 [60]	92.29	93.77	91.40	**93.09**	92.88	90.33	90.33
Hybrid workflows							
PaddleOCR-VL-1.6 [64]	**93.19**	**94.74**	92.48	92.78	**93.28**	**92.66**	**92.48**
Xiaomi-OCR-0 [217]	92.73	93.73	**92.76**	92.80	92.68	91.69	91.69
TeleOCR [63]	90.72	92.33	90.63	89.09	90.90	90.63	89.09
GLM-OCR [66]	90.32	92.67	90.68	91.75	91.12	85.39	85.39
MonkeyOCRv2-B-Parsing [77]	89.22	89.49	89.70	88.40	88.54	89.97	88.40
General-purpose VLMs							
Doubao-Seed-2.1-Pro [218]	89.02	88.85	89.36	88.99	89.13	88.79	88.79

CRNN to 97.7 for SVTRv2; and Total-Text F1 without a lexicon increased from 65.3 for Mask TextSpotter v2 to 84.6 for TiCLS (Tables 3–5). From the 2022 entries to the latest ones, the gains are 1.4, 0.7, and 1.0 points: most of the distance was covered within the Deep Learning Era, and the open problems moved to harder text distributions and to full pages.

Document structure components. Component benchmarks approached their ceilings next. Reported TEDS scores include 88.3 for EDD on the original PubTabNet test set [5] and 96.3 for TableMASTER on the ICDAR 2021 test split [219]; the different test sets prevent a direct estimate of improvement. On PubLayNet, reported mAP includes 91.0 for Mask R-CNN in the RoDLA evaluation [220] and 96.2 for VGT [221], and UniMERNet reached 96.8 CDM on UniMER-Test in a 2025 metric study [101]. These scores characterize individual components; full-page benchmarks additionally assess how their outputs preserve reading order and relations between elements.

VLM OCR diagnostics. Shared diagnostic suites measure the reading ability that general-purpose VLMs expose through prompts. Selected OCRBench results range from 736 for GPT-4o to 931 for Qwen3.5-397B-A17B [11], [222], an increase of 195 points that leaves the leading score 69 points below the nominal ceiling of 1000. The total aggregates 1,000 question-level judgments spanning text recognition, scene-text and document question answering, key information extraction, and handwritten mathematical expression recognition [11], so a change in the total cannot be attributed to any single ability, and reconstruction of complete pages lies outside the task set.

Parsing complete pages. Benchmarks for parsing complete pages evaluate reconstruction under a common output protocol. On OmniDocBench v1.6_full, the selected results span 93.26–96.91 Overall, 95.60–97.35 for text, 88.04–96.82 for tables, and 95.08–97.53 for formulas (Table 6) [60], [63], [155]. Text scores of the eight systems lie within 1.75 points of one another, and table scores spread over 8.78 points, so tables are the component that still separates them. TeleOCR and OvisOCR2 hold the top two positions, one from layout-guided regional generation and one from whole-page generation. Every system above 95 Overall has at most 1.2B parameters, while the 2B to 4B systems score at most 94.00. Within this group, the choice between the two designs does not decide the ranking, and neither does parameter count.

Physical robustness. Real5-OmniDocBench evaluates scanning, warping, screen photography, illumination change, and skew over the same 1,355 source pages from OmniDocBench v1.5. Across the selected systems on its current official leaderboard, overall scores span 64.45–93.19 (Table 7) [216]. The modular toolkit PP-StructureV3 falls from 84.68 on scans to 37.98 under skew, while every listed generative system stays above 85 in all five conditions, and their overall scores lie within 4.2 points of one another. Each image is a re-capture of one source page under a single acquisition condition [97], so documents that exist only as camera captures lie outside the benchmark's coverage.

4.2.2 What Current Metrics Miss

The trends above hold within fixed protocols. Across protocols, interpretation is limited by three recurring issues: differences in output definitions and matching rules, aggregation across heterogeneous components, and incomplete coverage of content types.

Ambiguity in output definitions and protocols. Document structure admits multiple reasonable decompositions and serializations. Borderless tables, aligned fields, forms, and layouts with multiple columns can cross category boundaries, while visually equivalent tables may have different HTML or LaTeX representations. Benchmarks also use different region ontologies, annotation granularities, and matching rules, as illustrated by PubLayNet, DocLayNet, and OmniDocBench [21], [96], [155], [179].

A reported score therefore reflects both the system's reconstruction and the protocol's definition of structure.

Composite score attribution. OmniDocBench v1.6 illustrates the attribution problem: its *Overall* score combines text, table, and formula metrics, while reading order is reported separately [155]. Similar overall scores can therefore represent different component profiles.

Coverage boundaries. Current benchmarks for complete pages concentrate on text, mathematical formulas, tables, layout regions, and reading order. Charts, flowcharts, chemical structures, and sheet music also encode content and structure that can be recovered from visual evidence; GOT, Logics-Parsing, and OCRVerse already output some of them from page images [8], [10], [223], and page-level benchmarks have begun to score charts [202].

5 Discussion and Outlook

This survey traces OCR from specialist recognition models through joint text spotting to unified document models. We define the task through faithful transcription, organize the current ecosystem into five coexisting system families, compare their architectures, data, and training, examine the limitations of generative parsing, and follow the evolution of benchmarks from isolated characters to complete documents. Looking ahead, three expectations stand out.

First, transcription is increasingly consumed directly by models and agents, and hallucinated or omitted content propagates downstream, so reliability becomes the primary requirement, judged against the source image. Second, requirements for accuracy, cost, and control differ across uses, so specialist, unified, and hybrid approaches will continue to coexist, with the cost of processing a page a primary constraint on that choice. Third, the object of reconstruction will keep widening, from text to the structure of whole documents, from single pages to many, from a few scripts to all writing systems, and from clean scans to whatever a camera captures, and evaluation must widen with it.

OCR's lasting value lies in turning visually recorded information into accurate digital content that can be searched, edited, processed, and reused.

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